COURSE CODE BTCE205	COURSE NAME DIGITAL ELECTRONICS	SEMESTER II

Teaching Scheme (Hours)				Teaching Credit			
Lecture	**Practical**	**Tutorial**	**Total Hours**	**Lecture**	**Practical**	**Tutorial**	**Total Credit**
3	2	0	5	3	1	0	4

Course Pre-requisites	Basics of electronics
Course Category	Basic Science Courses/ Engineering Science Courses/ Mandatory courses
Course focus	Skill development
Rationale	Digital Electronics holds significant relevance at the local, national, and international levels due to its pervasive impact on modern technology and communication systems. **Locally, understanding digital electronics enables individuals to operate and troubleshoot common consumer devices, such as smartphones, computers, and home appliances. Nationally, it is the backbone of various industries, including telecommunications, manufacturing, and information technology, driving economic growth and innovation. Internationally, digital electronics facilitates global connectivity, data exchange, and collaboration, enabling seamless communication and integration across borders.** Proficiency in digital electronics empowers individuals and nations to adapt to the rapidly evolving digital landscape, leverage emerging technologies, and contribute to the advancement of society in the digital age.
Course Revision/ Approval Date:	19/8/2019
Course Objectives (As per Blooms' Taxonomy)	To enable the student to: 1. To aware students about number system, 2. To familiar student with Boolean algebra 3. To make student understand regarding combinational circuit 4. To elaborate sequential circuits 5. To provide knowledge about memory devices.

Course Content (Theory)	Weightage	Contact hours
Unit 1: Data and number systems, Binary representation, Codes and their conversions: BCD, Octal, Hexadecimal, ASCII, EBDIC, Gray, Signed binary number representation with 1's and 2's complement methods.	**20%**	**9**
Unit 2: Binary arithmetic Boolean algebra, Venn diagram, logic gates and circuits, Minimization of logic expressions by algebraic method, Kmap method and QuineMcClauskey method	**20%**	**9**
Unit 3: Combinational circuits- adder, subtractor, encoder, decoder, comparator,multiplexer, de-multiplexer, parity generator, etc Design of combinational circuits- Programming logic devices and gate arrays	**20%**	**9**
Unit 4: Sequential Circuits- Flip Flops, various types of Registers and counters and their design, Irregular counter, State table and state transition diagram, sequential circuits design methodology	**20%**	**9**
Unit 5: Memory devices- ROM, RAM, EPROM, EEPROM, etc Different types of A/D and D/A conversion techniques Different Logic families- TTL, ECL, MOS and CMOS, their operation, design and specifications	**20%**	**9**

List Of Practical	Weightage	Contact hours
1: Study of Digital Number System & Its Significance. Study of Logic Gates (Buffer, AND, OR, NOT EXOR, EXNOR, NAND & NOR) Study of Adder circuit	**25%**	**6**
2: Study of Subtrator circuit Study of Parity Bit Generator Study of Sequential logic & Flip Flops	**25%**	**6**
3:	**25%**	**6**

Study of CMOS Technology & Its Significance in Digital Electronics		
4: Project Canvas	**25%**	**8**

Instructional Method and Pedagogy: (Max. 100 words) Visual Aids and Demonstrations, Hands-On Approach, Active Learning Strategies, Real-World Examples, Project-Based Learning, Continuous Assessment

Course Objectives:	**Blooms' Taxonomy Domain**	**Blooms' Taxonomy Sub Domain**
After successful completion of the above course, students will be able to: CO1: **Understand** the number system	Cognitive	Understand
CO2: **Apply** Boolean algebra for K-maps		Apply
CO3:**Analyze**combinational circuits		Analyse
CO4: **Understand** working of sequential circuits**.**		Understand
CO5:Comprehend **understanding** of memory structure		Understand

Learning Resources	
1.	Reference Books: 1. Morris Mano, Digital Logic and Computer Design 2. Anandkumar, Fundamental of Digital Circuits 3. R. P. Jain, Digital Electronics
2.	Video Reference: https://www.youtube.com/playlist?list=PLBlnK6fEyqRjMH3mWf6kwqiTbT98eAOm
3.	NPTEL MOOC https://nptel.ac.in/courses/117106086/

Evaluation Scheme	**Total Marks**
Theory: Mid semester Marks	20 marks
Theory: End Semester Marks	40 marks

Theory: Continuous Evaluation Component Marks	Attendance: 05 marks MCQs: 10 marks Open Book Assignment: 15 marks Open Book Assignment: 10 marks **Total: 40 Marks**
Practical Marks	Attendance: 05 marks Practical Exam: 20 marks Viva: 10 marks Journal: 10 marks Discipline: 05 marks **Total: 50 Marks**

Mapping of PSOs & Cos

	PSO1	PSO2	PSO2
CO1	2	3	2
CO2	2	3	1
CO3	1	2	1
CO4	2	1	1
CO5	1	1	3

Mapping of POs & Cos

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	0	1	0	0	0	0	0	1	1	2
CO2	3	1	0	2	0	0	0	0	0	1	1	1
CO3	3	2	1	1	0	0	0	0	0	2	1	3
CO4	3	0	0	1	0	0	0	0	0	0	1	2
CO5	3	1	3	1	0	0	0	0	0	2	1	2

1: Slight (low); 2: Moderate (Medium); 3: Substantial (High); 0 None